

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management is interested in purchasing patterns across customer demographics, product categories, subscription status, seasons, payment methods, and shipping preferences.

The analysis focuses on identifying the patterns that influence customer purchases and revenue performance, with particular attention to subscription adoption, gender-based sales, category revenue, seasonal performance, payment preferences, and shipping choices.

Overarching Business Question

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing, product, subscription, and service strategies?”

Key Business Objectives

1. Understand the overall customer base and purchasing behavior.
2. Measure subscription adoption and identify opportunities to convert non-subscribers.
3. Compare sales performance across male and female customers.
4. Identify high-performing product categories based on revenue.
5. Analyze seasonal revenue patterns to support marketing and inventory planning.
6. Understand customer preferences across payment methods.
7. Evaluate shipping type preferences and their relationship with purchase behavior.

Project Deliverables

No.	Deliverable	Description
1	Data Preparation & Modeling (Python)	Clean and transform the raw customer shopping dataset for analysis.
2	Data Analysis (SQL)	Structure the data and perform business-oriented analysis of customers, revenue, subscriptions, and purchasing behavior.
3	Visualization & Insights (Power BI)	Build an interactive dashboard showing KPIs, subscription status, revenue by category and season, sales by gender, payment methods, and shipping preferences.
4	Report & Presentation	Document the major findings, business insights, and actionable recommendations.
5	GitHub Repository	Maintain the Python notebook/scripts, SQL analysis, dashboard files, documentation, and supporting project assets in a structured repository.

Dashboard Alignment

Key dashboard KPIs: 3.9K customers | \$59.76 average purchase amount | 3.75 average review rating.

Dashboard analysis areas: subscription status, revenue by season, revenue by category, sales by gender, payment method split, and shipping type preference.